

Mixing diagnostics and reparameterisations for the polygenic PG-Gibbs sampler

Paul McKeigue

April 4, 2026

1 Model

We observe binary outcomes $y_i \sim \text{Bernoulli}(\sigma(\eta_i))$ for $i = 1, \dots, M$, with linear predictor

$$\eta_i = \phi + (LZ_{\text{mix}})_i - \mu \bar{m} v_i, \quad v = L\mathbf{1}, \quad \bar{m} = \tanh(\phi/2). \quad (1)$$

Here L is the block-diagonal Cholesky factor of the genetic relationship matrix (GRM), with $M = 10250$ individuals, $n_{\text{blocks}} = 8595$ blocks, and $n_{\text{relatives}} = 2658$.

Priors.

- $Z_{\text{mix},i} \mid r_i, \mu \sim \mathcal{N}(\mu r_i, 2\mu)$, where $r_i \in \{-1, +1\}$ with $P(r_i = +1) = p$.
- $p \sim \text{Beta}(K_\beta p_{\text{obs}}, K_\beta(1 - p_{\text{obs}}))$ with $K_\beta = 20$.
- $\mu \sim \text{half-Student-}t(\nu = 30, s = 1)$, sampled on the log scale $\theta = \log \mu$.

Polya-Gamma augmentation. Introducing $\omega_i \mid \eta_i \sim \text{PG}(1, |\eta_i|)$ renders the conditional distribution of Z_{mix} Gaussian. Writing $\kappa_i = y_i - 1/2$ and $\Omega = \text{diag}(\omega_i)$, the Gibbs cycle (implemented in `pg_gibbs_clean.py`) is:

1. Sample $\omega \mid \eta$ (Polya-Gamma draws).
2. Sample $Z_{\text{mix}} \mid \omega, \phi, \mu, r$ (Gaussian, block-by-block Cholesky solve).
3. Sample $r \mid Z_{\text{mix}}, \phi$ (independent Bernoulli per site).
4. Sample $\phi \mid \text{rest}$ (1-D slice sampling).
5. Sample $\mu \mid \omega, r, \phi$ collapsed over Z_{mix} (slice sampling; uses eigendecomposition of $L^\top \Omega L$ per block).
6. Refresh $Z_{\text{mix}} \mid \omega, \phi, \mu_{\text{new}}, r$.

The key quantity of interest is $\Lambda = \mu$, the expected log-likelihood ratio in nats (genetic information for discrimination).

2 Slow mixing: the hierarchical coupling

Observed effective sample size (ESS) from 2000 post-warmup samples with 8 chains is approximately 19–25, indicating severe autocorrelation. Two structural sources have been identified.

2.1 Centred-parameterisation coupling ($Z_{\text{mix}}-\theta$)

In the centred parameterisation, $Z_{\text{mix}} \mid r, \mu \sim \mathcal{N}(\mu r, 2\mu I)$. The conditional posterior $p(\theta \mid Z_{\text{mix}})$ is much narrower than the marginal $p(\theta)$: Z_{mix} is drawn at the current μ and then anchors the next θ update. Steps 5 and 6 alternate between a collapsed update of θ and a Z_{mix} refresh, but the class labels r are held fixed during the θ update, leaving residual coupling between Z_{mix} and θ .

2.2 ϕ - θ coupling through the prevalence constraint

The observed case rate $K = n_{\text{cases}}/M$ must be reproduced. Noting that the mean of η_i over individuals is approximately $\phi - \mu \bar{m} \bar{v}$ where $\bar{v} = M^{-1} \sum_i v_i$, the posterior concentrates near the ridge

$$K \approx \sigma(\phi - \mu \bar{m} \bar{v}), \quad (2)$$

giving approximately $\phi \approx \text{logit}(K) + \mu \bar{m} \bar{v}$. As μ increases, ϕ must increase proportionally to preserve the prevalence. In the RA dataset ($K \approx 0.25$) this yields a posterior correlation $r(\mu, \phi) \approx -0.985$.

3 ASIS: ancillarity-sufficiency interweaving

3.1 Construction

After step 5 (collapsed sufficient-statistic update for θ) and step 6 (Z_{mix} refresh at μ_{new}), define the non-centred parameterisation (NCP) ancillary statistic

$$u = \frac{Z_{\text{mix}} - \mu r}{\sqrt{2\mu}} \sim \mathcal{N}(0, I) \quad \text{under the prior.} \quad (3)$$

The ASIS step draws a fresh proposal for θ_{nc} from $p(\theta_{\text{nc}} \mid u, \omega, \phi, r, y)$ with u held fixed.

3.2 NCP log-posterior

With u fixed, the linear predictor becomes

$$\eta_i = \phi + \mu_{\text{nc}} a_i + \sqrt{2\mu_{\text{nc}}} b_i, \quad a = Lr - \bar{m} v, \quad b = Lu. \quad (4)$$

Because $p(u \mid \theta_{\text{nc}}) = \mathcal{N}(0, I)$ is independent of θ_{nc} , no Z_{mix} prior term appears in the log-posterior:

$$\log p_{\text{nc}}(\theta_{\text{nc}}) = \sum_i [\kappa_i \eta_i - \tfrac{1}{2} \omega_i \eta_i^2] + \theta_{\text{nc}} - \frac{\nu + 1}{2} \log \left(1 + \frac{\mu_{\text{nc}}^2}{\nu s^2} \right) + \text{const.} \quad (5)$$

This is sampled by 1-D slice sampling. After accepting θ_{nc}^* , the centred variable is reconstructed as $Z_{\text{mix}} = \mu^* r + \sqrt{2\mu^*} u$.

3.3 Effect

The interweaving exploits the fact that the centred and non-centred parameterisations have complementary geometry: the centred update mixes well when the signal is strong, the non-centred update mixes well when the signal is weak. Inserting the NCP step between steps 5 and 6 improves ESS from 19 to 25 and reduces $\hat{R}$ from 1.37 to 1.25 on the RA dataset.

4 Alpha reparameterisation for ϕ

4.1 Motivation

The ϕ - θ coupling identified in Section ?? (correlation ≈ -0.985) prevents the slice sampler from taking large steps in ϕ independently of μ . The remedy is to reparameterise in terms of the approximately orthogonal quantity

$$\alpha = \phi - \mu \bar{m} \bar{v} \approx \bar{\eta} \approx \text{logit}(K). \quad (6)$$

Since $\alpha \approx \text{logit}(K)$ throughout the posterior, α and θ are approximately orthogonal.

4.2 Implementation

Step 4 is replaced by a slice step for α at fixed μ and $\bar{m}_{\text{current}} = \tanh(\phi_{\text{current}}/2)$:

$$\phi^{\text{proposed}} = \alpha^{\text{proposed}} + \mu \bar{m}_{\text{current}} \bar{v}. \quad (7)$$

The Jacobian $\partial\phi/\partial\alpha = 1$, so no correction to the target density is needed. The log-posterior for α is simply the existing `logpost_phi_given_rest` evaluated at $\phi = \alpha + \text{shift}$:

$$\log p(\alpha) = \log p_\phi(\alpha + \mu \bar{m} \bar{v}). \quad (8)$$

After accepting α^* , the intercept is recovered as $\phi^* = \alpha^* + \mu \bar{m}_{\text{current}} \bar{v}$, and $\bar{m}$ is updated from ϕ^* for the next iteration.

In pseudocode:

```
mbar = tanh(phi / 2)
shift = mu * mbar * mean_v
alpha_current = phi - shift
# slice sample alpha from logpost_phi_given_rest(alpha + shift)
phi_new = alpha_new + shift
```

The reparameterisation is controlled by flag `use_alpha_reparam` (default `False`), preserving full backwards compatibility.

5 Results summary

Table ?? summarises ESS and convergence diagnostics across runs on the RA dataset (8 chains).

Table 1: Mixing diagnostics for different sampler configurations. n_w = number of warmup samples per chain; $\hat{\mu}$ and σ_μ are the posterior mean and standard deviation of μ .

Run	n_w	ESS	$\hat{R}$	$\hat{\mu}$	σ_μ
sw=0.5 fixed	1000	19	1.37	1.16	0.61
sw=1.0 fixed	500	23	1.29	1.05	0.54
sw=0.5 adapted (no phase 1)	2000	19	1.34	1.38	0.59
sw=0.5 ASIS	2000	25	1.25	1.35	0.56
sw=0.5 adapted (phase 1 = 4000)	5000	60	1.12	1.28	0.60

The largest improvement came from sufficient warmup: $n_w = 5000$ with a 4000-iteration phase 1 achieved ESS = 60 and $\hat{R} = 1.12$. When adaptation started too early (no phase 1), the adapted slice width collapsed to 0.033; with adequate phase 1 warmup it settled at a healthy 0.050. The algorithmic changes (ASIS, α -reparameterisation) provide moderate incremental gains but do not substitute for adequate warmup in this regime.